

Systems Modelling and Simulation

1) Discrete random variables

Consider an experiment where a fair coin is tossed once. Define a random variable X such that $X=1$ if the outcome is **Heads**, and $X=0$ if the outcome is **Tails**. Since the coin is fair, the probability of obtaining Heads is 0.5 and the probability of obtaining Tails is also 0.5.

- a) Find the **expected value** $E[X]$
- b) Find the **variance** $\text{Var}(X)$

2) Continuous random variables

- c) Explain what a continuous random variable is.
- d) How it differs from a discrete random variable.
- e) Discuss the idea of probability being assigned to intervals instead of specific values and explain the role of a probability density function (pdf) in continuous probability.
- f) Why integration is used to calculate mean values and variance.
- g) Finally, choose one real-life example of a continuous random variable calculate mean values and variance.